

Quade Butler

RESEARCH ENGINEER | STATE ESTIMATION, SENSOR FUSION, ROBOTICS

(705) 716-0431 | quadedbutler@gmail.com | quade-butler.github.io | linkedin.com/in/quadebutler

Experience

Research Assistant, McMaster University – Hamilton, ON

Jan 2026 – Present

- Prototyped recursive estimation algorithms in Python for nonlinear sensor data, evaluating accuracy, runtime, and robustness under measurement noise, outliers, and model uncertainty
- Built robust filters using dynamic measurement residual saturation to reject sensor outliers, improving robustness 12%
- Decreased filter runtime by improving point coverage for numerical integration, reducing function evaluations by 60% while preserving accuracy for high-dimensional systems
- Profiled filter performance to find execution bottlenecks, leading to improved processing time and numerical stability

PhD Researcher, McMaster University – Hamilton, ON

Jan 2021 – Dec 2025

- Developed algorithms for state estimation (Kalman filters) and multiple-model fault detection for MR damper devices
- Constructed cubature Kalman filters (CKF) by exploiting properties of the trapezoid rule, reducing computational cost 100x over vanilla Monte Carlo and Gauss-Hermite integration methods for high-dimensional functions
- Applied filter to radar tracking in MATLAB, reducing error by 50% versus UKF/CKF and achieving fewer lost target events
- Calibrated mode-dependent MR damper models using constrained Gauss-Newton optimization across 60 datasets sampled at 100 Hz, validated using 80-20 train-test split and evaluating damper hysteresis envelopes
- Architected an interacting multiple model sensor fusion estimator for force estimation and fault detection/classification of an experimental MR damper, achieving 99.2% accuracy and 99.1% macro F1 for induced fault regimes

Research Collaboration with Ford Motor Company – Livonia, MI

May 2020 – Aug 2024

- Led the design and development of a condition monitoring system for machine tools using built-in sensors, reducing unplanned downtime and avoiding both costs from third-party solutions and the need for external sensor systems
- Designed simulated motion trajectories in MATLAB and deployed them on production machines to generate repeatable sensor datasets for condition monitoring validation
- Built a data processing and analysis pipeline in Python to transform, clean, and extract features from noisy XML datasets
- Transformed raw sensor data into useful machine health indicators, enabling the early detection of gradual wear failures

Projects

Custom C++ Machine Learning (ML) Library

- Implemented C++ image classification library with binary data ingestion, matrix operations, backpropagation with automatic differentiation, and cross-entropy training; achieved 96% test accuracy on MNIST digits for single layer network
- Built custom unit tests for matrix types, validating shape handling, indexing behavior, dimension mismatch errors, arithmetic correctness, and edge-case behavior for numerical ML routines
- Evaluated loop-order optimizations for matrix multiplication by comparing against naïve implementations, decreasing training times by 2-4x without the use of external ML libraries or GPU acceleration

Skills

State Estimation & Robotics: Kalman filtering, state estimation, localization, sensor fusion, interacting multiple models

Languages & Tools: Python, C++, MATLAB, NumPy, SciPy, git, CMake, Linux

Education

Ph.D., Mechanical Engineering, McMaster University

Ontario Graduate Scholar

Dec 2025

B.Eng., Mechanical Engineering, University of Guelph

Minor in Mathematics

2020